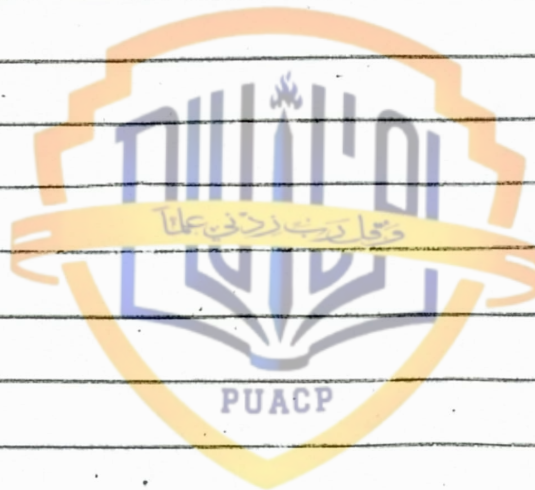


Public Sector Economics

8th Sem

04-02-2025



Introduction

Definition: →

The branch of economics that studies the role of the govt in the economy.

Focus areas: →

- Public goods, govt expenditure
- Taxation and Public policy.

Importance: →

Understanding how govt actions influence economic efficiency and equity.

Concept of Budget: →

A financial plan that outlines the govt anticipated revenues and expenditure over a period of time.

Components: →

- Revenues
- Expenditures.

Fiscal year: →

A fiscal year is one year period used by governments and organizations for budgeting, accounting, and financial reporting purposes, which may or may not align with the calendar year.

FY of Pakistan: → July 1st — 30 June
FY of US → Oct 1 — Sep 30

Objectives of Budget: →

- Resource allocation
- Stabilization
- Distribution
- Public Accountability.

Types of Budget: →

- Budget deficit
- Budget surplus
- Balanced Budget

Public Expenditures	Private Expenditures
Funded by govt	Funded by individual
Providing public goods	Personal gain
Economic stability and social welfare	Personal preferences and market condition

Main Source of Public Revenue: →

- Taxes:
 - Direct taxes
 - Indirect taxes
- Fees and charges: →
- Borrowings
- Grants and Aids.

←: Public Expenditure: →

Public expenditure refers to the govt spending on goods and services to achieve economic and social objectives:

Objectives : —————> Importance : —————>

- > Provide public goods and services
- > Redistributes income
- > stabilize economy
- > Promote economic growth

Causes of Public Expenditure Growth

- > Economic growth
- > Urbanization and industrialization
- > Aging population
- > Healthcare advancements.
- > Education expansion.
- > Infrastructure development
- > Social welfare programs
- > National security concerns
- > Government intervention in economy.
- > Technological advancement
- > Trade liberalization

Major Theories of Public expenditures : —————>

Wagner's law : —————>

Proposed by Adolph Wagner in 1883, Posits that as an economy expands, Public expenditure tends to grow at a faster rate than national income.

According to Wagner the growth in public expenditures over time should be greater than that of national output.

Reason for growth : —————>

- > increased demand for services : —————>

Modern societies requires government intervention. I.e. to correct market failures.

Social and Economic change: →

Urbanization, industrialization

Population growth: →

Implications: →

Fiscal Policy: →

governments may need to adjust their fiscal policies to align with increasing public spending requirements.

Taxation: →

Rising public expenditures may need to adjust their fiscal policies for higher tax rate on high level of income.

Economic management: →

effective economic planning and public finance management.

Critiques and Limitations: →

→ Applicability: →

→ External factors: →

such as political decisions, Economic Crises and Technological advancements.

Colin Clark Hypothesis: →

Colin Clark introduced the hypothesis in 1945 as a part of his research on the relationship between the public expenditures

and economic performance.

The hypothesis emerged in post world war II environment where many governments were expanding their role in economic recovery and social welfare.

The Colin Clark hypothesis also known as "critical limit hypothesis" posits that government expenditures should not exceed 25% of country's Gross national product (GNP) or (GDP) to maintain optimal economic efficiency.

Beyond this threshold, excessive government spending can lead to diminishing returns, inefficiency and potential economic stagnation.

Optimal Range for developing countries

Clark suggested an optimal government spending level between 15% and 25% of GNP / GDP

Consequences of Excessive Spendings:

- Diminishing Returns
- Inefficiency
- Economic stagnation

Market Failure

Market failure occurs when the free market, operating on its own, does not allocate resource efficiently.

This inefficiency results in a loss of economic and social welfare.

When market fails, the private incentives of individuals or firms do not align with societies overall interests, leading to outcomes that are suboptimal.

Market failure occurs when the distribution of goods and services in a free market is inefficient, leading to economic value, it can happen when the quantity of goods or services supplied is not equal of the quantity demanded.

Market failure refers to a situation defined by an inefficient distribution of goods and services in the free market.

Causes of Market failure: →

- Externalities.

These occur when someone's actions effect others who aren't involved in the decision.

- Positive externalities

- Negative externalities

- Public goods

The goods which people uses without paying for them and use of the good by one person does

not reduce availability for others.

→ Imperfect information / information asymmetry: →

→ Market power: →

some companies have enough control over a market to set prices or limit production.

→ inequality: →

→ Transaction cost: →

High costs associated with making exchanges can prevent beneficial trades.

→ Business cycle: →

Economic ups and downs can lead to wasted resources.

Externality: →

An externality refers to the cost or benefits of an activity that is experienced by unrelated third party who are not directly involved in the activity.

Examples:

→ Smoking impact on other's

→ Pollution, waste material.

Negative externality: →

Positive externality: →

Consumption externality: →

Production externality: →

Internalizing externalities: →

The process of incorporating external costs or benefits into the decision making of individuals or

firms, aligning private incentives with social welfare.

Importance: →

→ Correct market failures by ensuring that the true costs and benefits of goods and services are reflected in their prices.

→ Promote efficient resource allocation and enhance overall societal well-being.

Methods of internalization: →

→ Pigouvian taxes: →

Taxes imposed on activities that generate negative externalities.

For example: →

Carbon emission tax

→ Subsidies for positive externalities: →

Financial support for activities that create positive externalities.

→ Regulations

→ Tradable permits

Allow firms for buying and selling permits for pollution, create a market for externality reduction.

→ Liability and legal action.

Holding firms accountable for external costs through legal means.

Benefits of internalizing externality: →

→ Efficiency

→ Social welfare

→ Innovation

Important terms: →

→ Marginal Private Cost (MPC)

- Marginal external cost
- Marginal social cost

$$MSC = MPC + MEC$$

- Marginal Private benefit
- Marginal External benefit
- Marginal social benefit.

$$MSB = MPB + MEB.$$

Socially optimal output : →

SoO refers to the level of production or consumption that maximizes overall social welfare. This concept is especially relevant in the context of externalities, where the private market outcome may not align with what is best for society as a whole.

Definition : →

The quantity of goods or services produced or consumed that leads to the highest net benefit for society. This occurs when the marginal social cost (MSC) of producing a good equals the marginal social benefit (MSB) derived from its consumption.

Finding socially optimal output

The socially optimal output is achieved when

$$MSC = MSB$$

At this point, the net

benefit for the society is maximized because the cost of producing one more units equals the value that society place on that unit.

~~Coase's~~

Coase theorem

Coase theorem was represented by Ronald Coase in his most influential paper, 'The problem of social cost' (1960), he developed what later became known as the Coase theorem.

Theorem : →

The Coase theorem is a key concept in economic theory that addresses how to deal with externalities — cost or benefits that affect third parties not directly involved in a transaction.

The theorem provides insights into how private parties can negotiate solutions to externalities with government intervention.

Assumptions : →

- well defined property rights
- Zero transaction/negotiation cost
- Efficient outcome

Implication : →

- Efficiency without government intervention
- Role of transaction cost
- Policy implications

Limitations: →

- High transaction costs
- Unclear property rights
- Disparities in bargaining power
- No money consideration

Public Goods

Public goods are goods that are available for everyone to consume regardless of who pays for them. These goods are characterized by two main properties

- non-excludability
- non-rivalrous

Types of goods

- Private goods: →
- Public goods: →
- Common goods: →

Common goods are those that are non-excludable, but rivalrous, in other words, consumers cannot be stopped from benefiting from the good. However, it is rivalrous, which means that some may be excluded due to the consumption of others.

Example: →

Fish stocks, coal, timber

- club goods: →

club goods are those that are

excludable

In other words, a private firm is able to prevent people who do not pay from using it. Yet its increased consumption does not prevent others from accessing it.

Example: →

TV, parks, magazine

The Free-Rider Problem: →

The free rider problem occurs when individuals benefit from a public good or service without contributing to its cost, potentially leading to under provision or non-provision of the good.

Asymmetric Information: →

Asymmetric information refers to the situation where one party in a transaction has more or better information than the other. This imbalance can lead to inefficiency and problems in market.

Adverse Selection: →

Adverse selection occurs when one side of the market has information that the other side lacks, leading to a selection of poor quality product or higher-risk customers.

Moral hazard

Hidden action / behaviour after agreement.

Moral hazards arises when a party is more likely to take risks because they don't bear the full consequences of those risks, often due to lack of monitoring or accountability.

Example of car insurance.

Consequences: →

- Market failure
- Inefficient allocation
- Reduced Trade
- Unfair outcomes

←: Tax System: →

A tax system is a structured set of rules, policies and institutions established by the govt to collect revenue from individuals, businesses and other entities to fund public goods, services and programs. It involves a framework for determining the types of taxes, tax rates and methods for tax collection and enforcement. The tax system plays a vital role in the economy by financing govt activities, promoting economic stability and redistributing wealth.

Type of Taxes:

- Income tax
- Sales tax

- Property tax
- Excise tax

Tax Base

The amount on which a tax is imposed, such as income, property value. The tax base is fundamental to understanding how much tax a person or business will pay under a given tax system.

Tax Rates: →

- Progressive tax
- Regressive tax
- Flat tax

Tax Collection methods

① → withholding tax: →

A system where tax is directly deducted from a taxpayer's income or transaction at the source, such as through payroll.

② → Self Assessment: →

Taxpayers calculate and submit their own tax liabilities, typical for income tax or business profits.

③ → Indirect tax: →

Tax Administration: →

The govt agencies responsible for enforcing tax laws and ensuring compliance.

- US (IRS) Internal Revenue Service
- UK (HMRC) His Majesty Revenue and Customs
- PAK (FBR) Federal Board of Revenue.

Types of tax systems:

- Progressive tax system
- Regressive tax system
- Proportional tax system
- Indirect tax system
- Direct tax system
- Capital Gain tax system
- Hybrid tax system.

Pakistan tax system

Pakistan's tax system is a blend of direct and indirect taxes, aimed at funding government operations and promoting development.

Key Components:

① → Income tax:

Personal Income tax:

Progressive tax rates ranging from 2.5% to 35% (35% for incomes over 60lac.) (Budget 2024)

Corporate Income tax:

Tax rate set at 29% for most companies (Budget 2024).

② → Sales GST:

17% standard sales tax is applied.

③ → Capital gain tax:

Capital gain tax for filers on securities is 15% and for non-filers it is 45%.

④ withholding tax: →

various payment such as dividends, salaries, and interest are the subjects to withholding tax.

Desireable characteristics of Tax system: →

- Equity
- Efficiency
- Simplicity
- Flexibility
- Certainty and stability
- Revenue sufficiency
- Transparency and Accountability
- Compliance



ad valorem tax is a tax based on the price of the product. It is usually a Percentage of the price.
Example: → VAT 10% on a 100\$ item.

Impact of tax refers to who is legally responsible for paying the tax (The business or consumer)

Tax incidence: →

refers to who actually bears the cost of the tax depending on market factors such as elasticity.

who bears the cost (consumer or producer)

Specific tax incidence: →

This refers to the distribution of the burden of a specific tax, based on the price elasticity of supply and demand.

If the supply of good is inelastic, producer will bear more of the burden of a specific tax.

Differential tax incidence: →

Examines that how different individuals or groups are affected by a tax, depending on factors like income or consumption patterns.

Luxury goods taxes may place a heavier burden on wealthy individuals.

Balanced budget incidence: →

This refers to the idea that if a government imposes tax, the additional revenue generated should be used to balance the government budget, ensuring the tax burden is spread evenly and efficiently across different sectors.

Theories of taxation

Cost of service Principle: →

Adam Smith (wealth of nations)
Tax should reflect the cost of government services provided to individuals. People pay in proportion to the

services they use.

Example: →

Toll tax on roads.

Benefit Principle: →

Knut Wicksell and Erick Lindahl

Tax should be based on the benefits received from government services. The more a person benefits, the more they should pay.

Example: → Road Tax

Ability to Pay Principle: →

Arthur Cecil Pigou: →

Tax should be levied based on an individual's ability to pay, regardless of the benefits they receive. This statement supports progressive taxation.

Approaches to taxation: →

(1) → objective tax approach: →

This approach is based on facts and verifiable evidence and is unbiased and impartial.

(2) → subjective tax approach: →

This approach is based on emotions, opinions and feelings.

Tax incidence: →

The division of tax burden between stakeholders, such as buyers and sellers or producers.

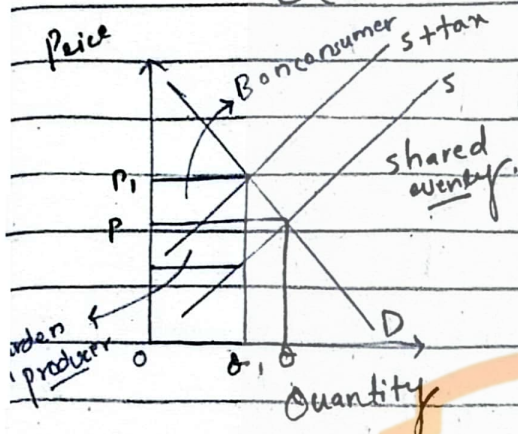
and consumers.

Formula for consumer tax burden: $\rightarrow$

$$E(\text{supply}) / E(\text{demand}) + E(\text{supply})$$

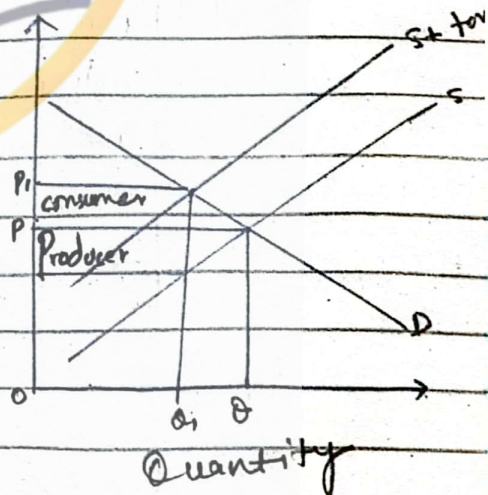
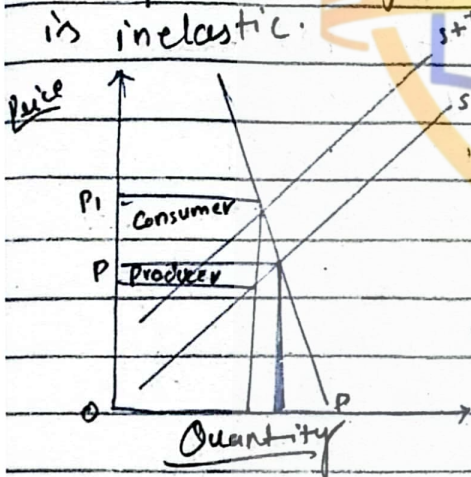
Formula for producers tax burden: $\rightarrow$

$$E(\text{demand}) / E(\text{demand}) + E(\text{supply})$$



Consumer shared more burden when price elasticity demand is inelastic.

Produce shared more burden when PED is elastic.



Optimal Tax rate

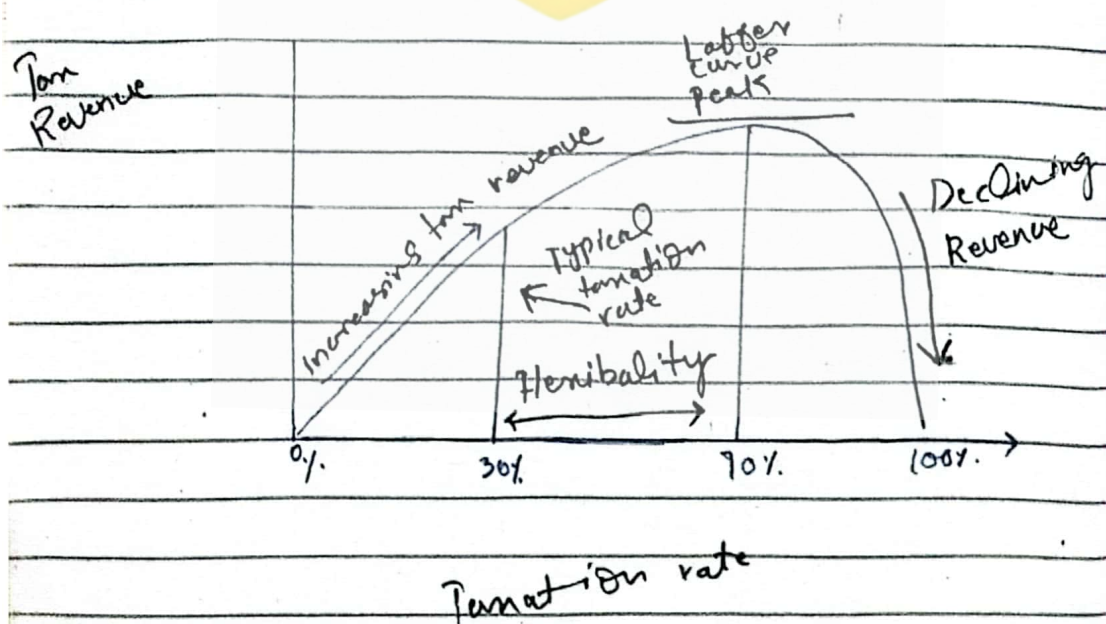
The optimal tax rate that generates the most benefit for the society by rising enough revenue for government services, while causing at least negative impact on people's economic

decisions and behaviour. It balances fairness and efficiency in the tax system.

Laffer Curve : $\Rightarrow$

The Laffer Curve is a theoretical concept in economics that proposes a relationship between tax rates and the resulting govt revenue. It's often depicted as an inverted U-shaped curve, suggesting that:

- $\Rightarrow$ At 0% tax rate there's no revenue
- $\Rightarrow$ As tax rates increase, revenue also increases, but at a decreasing rate.
- $\Rightarrow$ At a certain "sweet spot" tax rate revenue reaches Maximum.
- $\Rightarrow$ Beyond that point tax increase will lead to decline in the Revenue.
- $\Rightarrow$ Ultimately, at a 100% tax rate, all economic activity would cease, leading to zero revenue.



Cost benefit Analysis

Private Cost: →

refers to the direct expenses incurred by an individual, business or organization in producing or consuming a good or service, including;

- Labour cost → Raw materials
- Capital stock costs → Marketing expenses
- Operating costs.

Social Cost: →

Social Cost refers to the indirect or negative consequences born by society as a whole, resulting from a consumption or production of good and service.

- Environmental degradation
- Health impacts
- Social Unrest
- Congestion

Types of costs: →

- Fixed cost → Variable Cost
- Direct cost → Indirect cost
- opportunity cost → Sunk cost
- Social cost → Private cost
- Intangible cost → Implicit Cost

Types of benefits: →

- Tangible benefits (quantifiable)
- Intangible benefits (non-quantifiable)
- Financial benefits (Monetary)
- Non-financial benefits (non-monetary)
- Social benefits (social impact)

- Private benefits (individual gain)
- Economic benefits (market impact)
- Strategic benefits (long-term benefits)

CBA (Cost and Benefit Analysis)

Cost-benefit analysis is a decision-making tool that compares the costs of a project, policy or action to its benefits, to determine whether it's worthwhile.

Consider's the following Criteria: →

- Relevance
- Feasibility
- Efficiency
- Effectiveness
- Equity
- Sustainability
- Transparency

Project Evaluation: →

A systematic process of assessing a project's feasibility, effectiveness, efficiency and impact.

Purpose: →

- Determine the viability of a project
- Assess potential benefits and costs
- Guide decisions-making for resource allocation

Phases: →

Pre-Evaluation: →

- Define objectives and scope
- Identify stakeholders
- Collect baseline data

Mid Evaluation : →

- Monitor progress
- Assess interim performance
- Adjust strategies if necessary

Post Evaluation : →

- Measures outcome against objectives
- Assess sustainability and long-term impact.
- Derive lessons for future objects.

Types of Project Evaluation

* → Ex-Ante Evaluation : →

- Conducted before implementation
- Focus: Feasibility, cost and benefit analysis, and risk assessment.

* → Ongoing / Midterm Evaluation : →

- Performed during the project cycle
- Focus: Monitoring progress and identifying improvements.

* → Ex-Post Evaluation : →

- Conducted after the project completion.
- Focus: Measuring outcome, impact, and sustainability.

Economic tools for Project Evaluation : →

→ (CBA)

- Compare total cost with total benefits

Key Metrics

• NPV

$$\geq \frac{B_t - C_t}{(1+r)^t}$$

Present value of Benefits

• Benefit cost ratio (BCR)

Present value of Costs

• IRR

The discount rate where NPV = 0

Decision Rule : Accept projects where